

ECE 3337: Signals & Systems Analysis

Lecture 22: Fourier Analysis Techniques

Reference: Chapter 5

(Section 5.11 - 5.12, some examples from Chapter 6)

Please turn off your gadgets now



Announcements

- Dr. Roysam will be on travel week of April 20th
- Classes will be held online by Dr. Reddy
- The Zoom link is below:

<https://uh-edu-cougarnet.zoom.us/j/89206792357?pwd=MdV6UBmjZ2jnmaXramZRQmpLWHEdNw.1>

Meeting ID: 892 0679 2357

Passcode: 576641

- Make sure to pickup your graded exams today
 - *Deadline for Regrade requests is Friday 5pm*



Final Exam

Date: May 11th 2 - 4pm, SW 102 (Social Work Building).

<https://www.uh.edu/infotech/services/facilities-equipment/supported-classrooms/sw/102/>

Plenty of old exams are posted on MS Teams for practice

- Practice for accuracy and speed
- Test yourself individually, rather than as a group
- Avoid a false sense of knowing a topic
 - This can easily happen in a group setting
- Suggested study strategy:
 - Do the practice tests individually
 - Identify the challenging problems, and ask your instructors for help
 - Discuss the challenging problems in your study group
 - Follow up with individual work to confirm your understanding



Recap: The Fourier Transform



$$\hat{\mathbf{X}}(\omega) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

**SUITABLE FOR
NON-PERIODIC
WAVEFORMS**

$$x(t) = \mathcal{F}^{-1}[\hat{\mathbf{X}}(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{\mathbf{X}}(\omega) e^{j\omega t} d\omega$$

$$x(t) \leftrightarrow \hat{\mathbf{X}}(\omega)$$

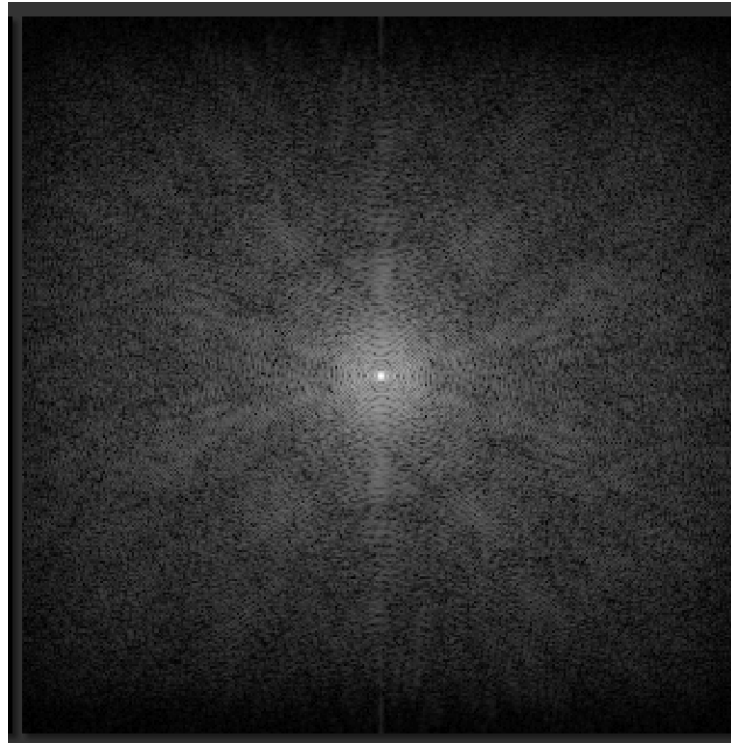
**The hat symbol ^ over $\mathbf{X}$ distinguishes this from the
Laplace Transform**



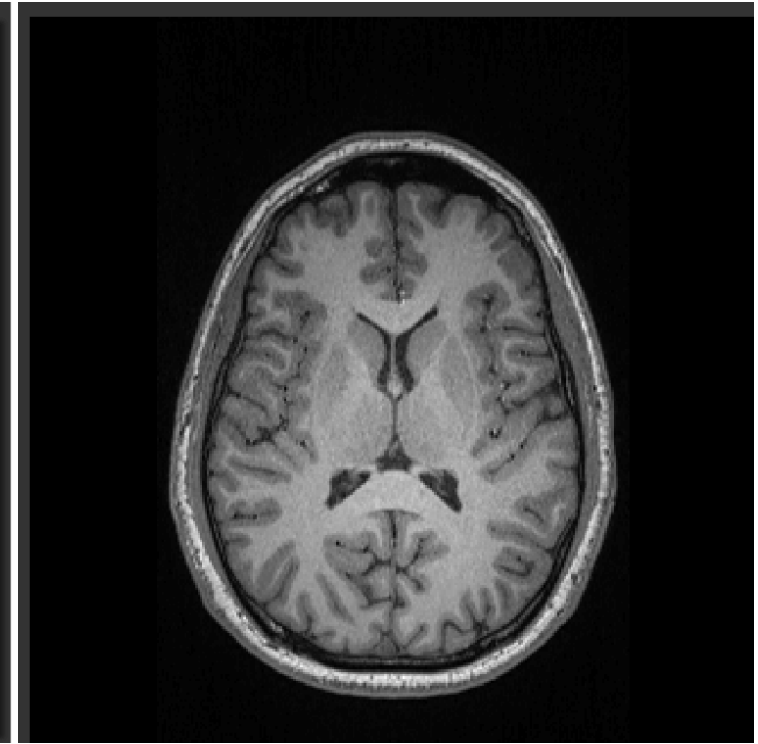
The Fourier Transform in Action: MRI



INSTRUMENT



DATA



COMPUTED IMAGE

In Magnetic Resonance Imaging (MRI), the instrument collects the (two-dimensional) Fourier transform of the brain image.

Computing the inverse Fourier transform produces the brain image on the right



Caveat

Not every function has a Fourier transform !

- The Fourier integral needs to converge for the transform to exist !
- Generally, the transform exists if the signal has no discontinuities, and is absolutely integrable

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty$$

- Luckily, some discontinuous signals of interest to us (delta, step) nevertheless have Fourier transforms, but we'll need some tricks to calculate them



Relating to the Laplace Transform

Recall the Laplace transform:

$$\mathbf{X(s)} = \int\limits_{\substack{+\infty \\ 0-}} x(t)e^{-st} dt = \mathcal{L}[x(t)] \quad \text{Unilateral} \\ \text{(one-sided)}$$

$$\mathbf{X(s)} = \int\limits_{\substack{+\infty \\ -\infty}} x(t)e^{-st} dt = \mathcal{L}[x(t)] \quad \text{Bilateral} \\ \text{(two-sided)}$$

$$\mathbf{s} = \sigma + j\omega \quad \text{A complex number}$$

The Fourier Transform can be viewed as a special case of the bilateral (two-sided) Laplace Transform when $\sigma = 0$



Fourier Transform Properties

The properties of the Fourier transform are very similar to those of the Laplace transform

- Keep in mind that we mostly studied the unilateral Laplace transform
- Example: Linearity

$$x_1(t) \leftrightarrow \hat{X}_1(\omega)$$

$$x_2(t) \leftrightarrow \hat{X}_2(\omega)$$

$$\Rightarrow K_1 x_1(t) + K_2 x_2(t) \leftrightarrow K_1 \hat{X}_1(\omega) + K_2 \hat{X}_2(\omega)$$



The Scaling Property

$$\mathcal{F}[x(at)] = \hat{\mathbf{X}}(\omega) = \int_{-\infty}^{\infty} x(at) e^{-j\omega t} dt$$

$$\text{Let } u = at \quad \Rightarrow du = a dt$$

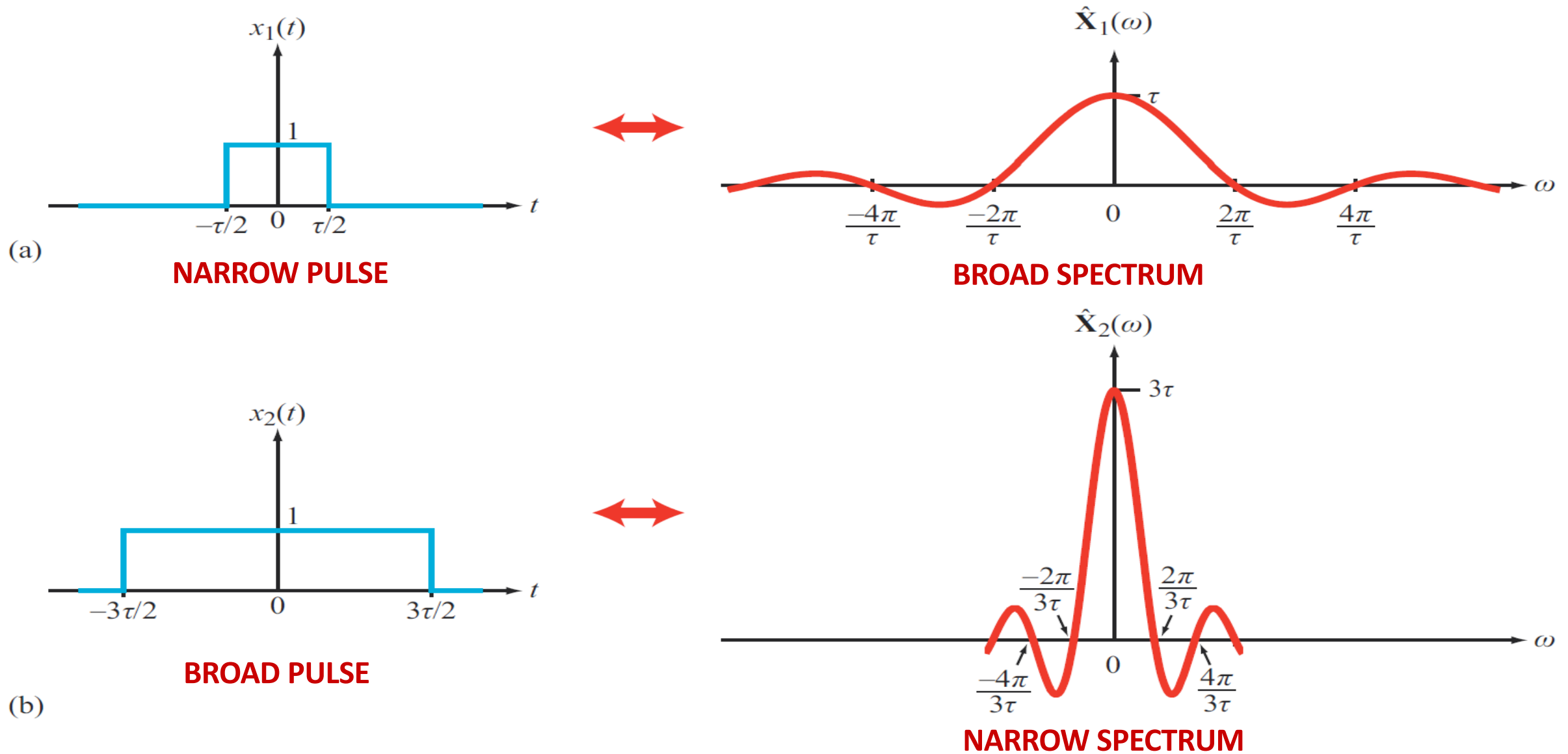
$$= \int_{-\infty}^{\infty} x(u) e^{\frac{-j\omega u}{a}} \frac{1}{a} dt = \frac{1}{a} \hat{\mathbf{X}}\left(\frac{\omega}{a}\right)$$

$$\Rightarrow x(at) \leftrightarrow \frac{1}{|a|} \hat{\mathbf{X}}\left(\frac{\omega}{a}\right)$$

**SIMILAR TO
LAPLACE
TRANSFORM**



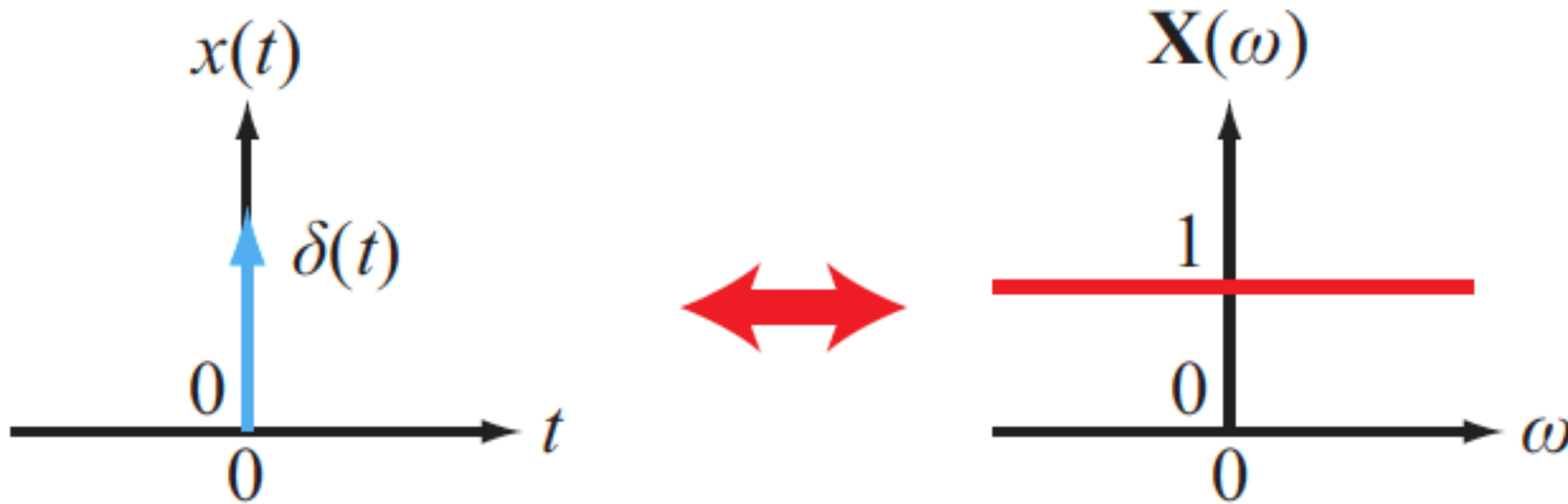
Recap: Time Scaling





Extreme Case: The Impulse

$$\mathcal{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} \delta(t) e^0 dt = 1 \quad \text{SNAPSHOT}$$



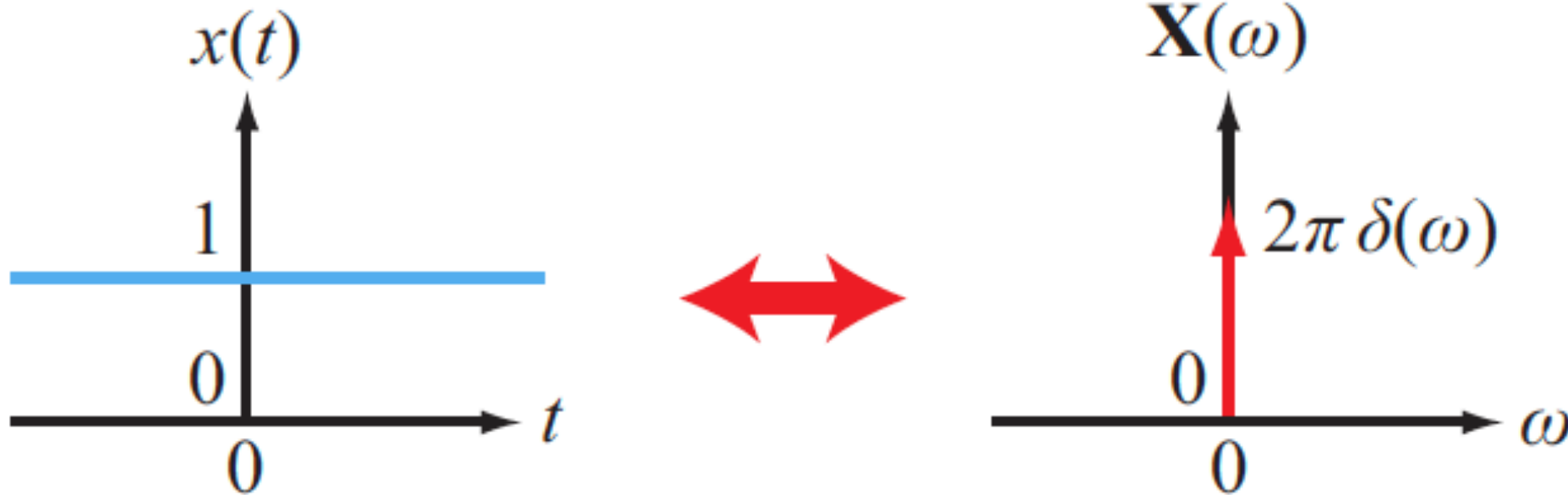
EXTREME CASE OF
NARROW PULSE $\longleftrightarrow$ BROAD SPECTRUM



The Impulse in Frequency Space

$$\mathcal{F}^{-1}[\delta(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) e^0 d\omega = \frac{1}{2\pi}$$

SNAPSHOT



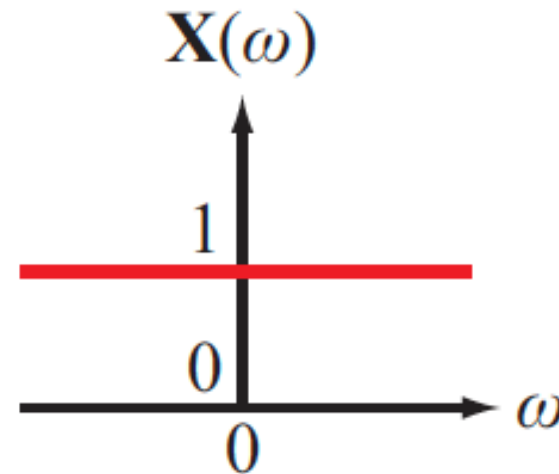
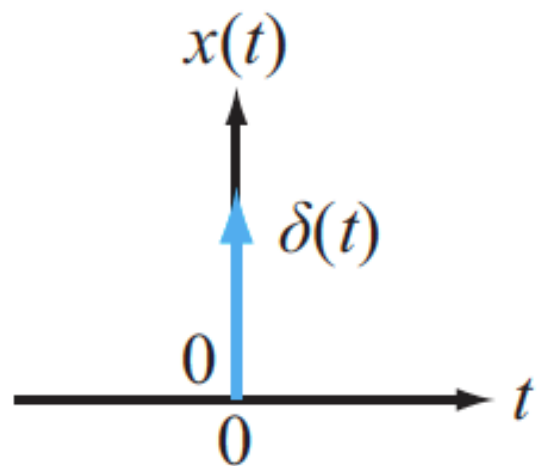
**EXTREME CASE OF
WIDE PULSE $\longleftrightarrow$ NARROW SPECTRUM**



Duality

$$\delta(t) \longleftrightarrow 1.$$

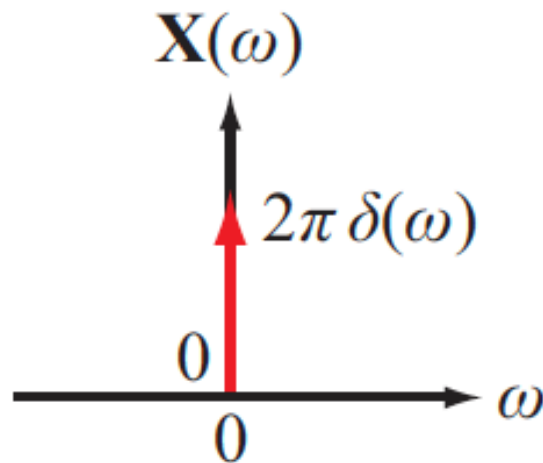
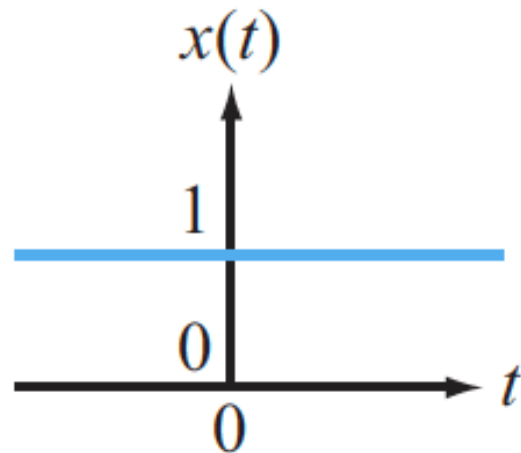
$$1 \longleftrightarrow 2\pi \delta(\omega).$$



THIS DUALITY IS WORTH REMEMBERING

AN ULTRA-NARROW PULSE IS ALSO CALLED A "BROADBAND PULSE"

(THIS IS A CASE OF ENGINEERS SEEMINGLY MIXING UP TIME AND FREQUENCY DOMAINS)





Decaying Exponentials

Let's try to calculate the Fourier Transform of the familiar decaying exponential:

$$x(t) = e^{-at} u(t) \quad a > 0$$

$$\mathcal{F}[x(t)] = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j\omega t} dt = \int_0^{\infty} e^{-at} e^{-j\omega t} dt = \left(\frac{-1}{(a + j\omega)} e^{-(a + j\omega)t} \right) \bigg|_{t=0}^{t \rightarrow \infty}$$

$$= \frac{1}{a + j\omega}$$

**SIMILAR TO LAPLACE
TRANSFORM ($s \sim j\omega$)**



The Unit Step

$$x(t) = u(t)$$

$$\mathcal{F}[x(t)] = \int_0^{\infty} u(t) e^{-j\omega t} dt$$

$$= \left(\frac{1}{-j\omega} \right) e^{-j\omega t} \Big|_0^{\infty} = \frac{j}{\omega} \left[e^{-j\infty} - 1 \right]$$

**WE NEED TO
FIGURE OUT AN
ALTERNATIVE WAY**

**YIKES!
DOES NOT CONVERGE
(INDETERMINATE)**



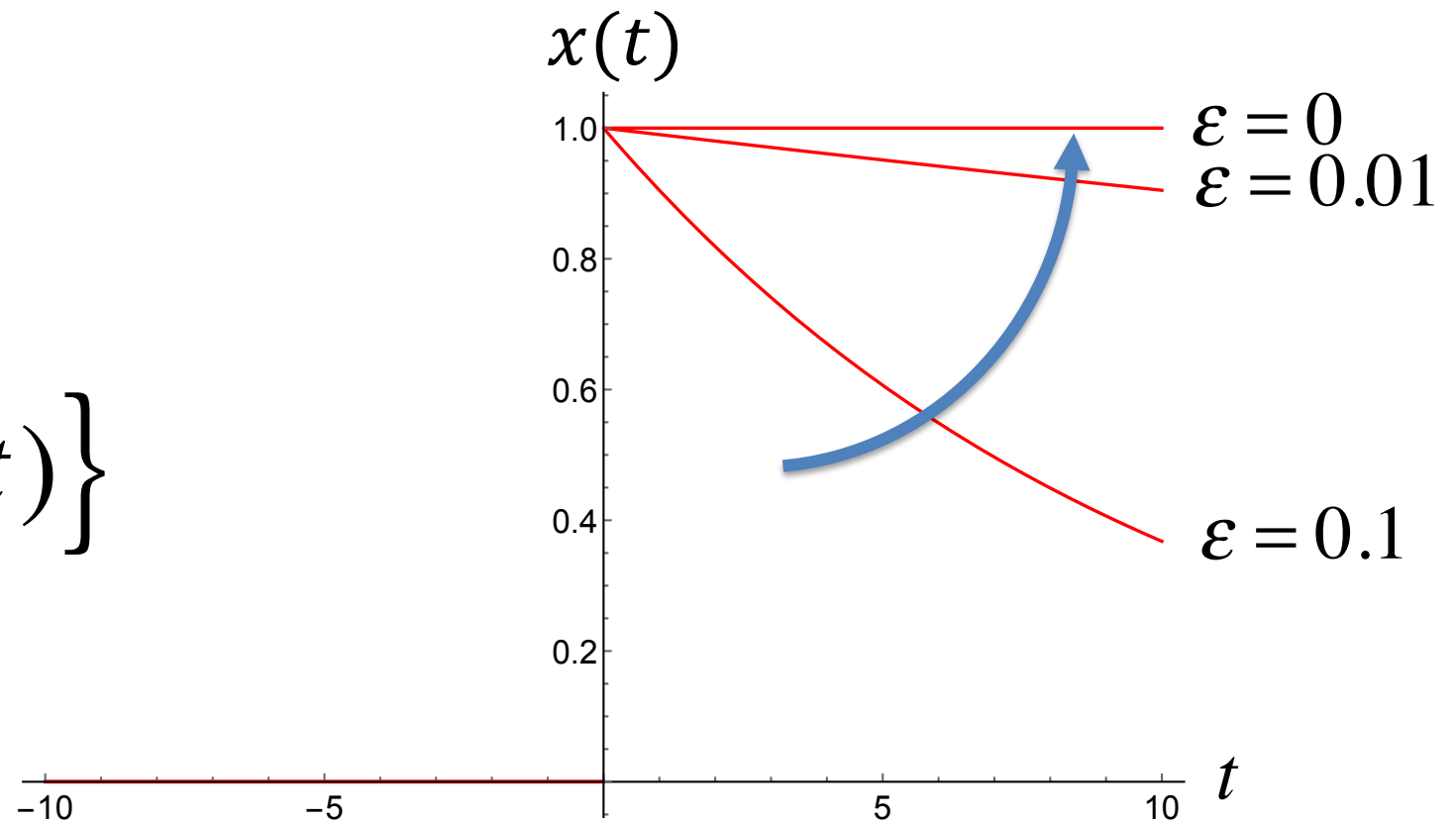
Two Approaches

There are two approaches to ‘tiptoe’ around:

#1. Write the unit step as the limiting case of the decaying exponential as $\varepsilon \rightarrow 0$

$$x(t) = e^{-\varepsilon t} u(t)$$

$$u(t) = \lim_{\varepsilon \rightarrow 0} \left\{ e^{-\varepsilon t} u(t) \right\}$$





First Approach

$$\mathcal{F}\left[e^{-\varepsilon t}u(t)\right] = \frac{1}{\varepsilon + j\omega},$$

$$\begin{aligned}\mathcal{F}\left[u(t)\right] &= \lim_{\varepsilon \rightarrow 0} \left\{ \frac{1}{\varepsilon + j\omega} \right\} = \lim_{\varepsilon \rightarrow 0} \left\{ \frac{(\varepsilon - j\omega)}{(\varepsilon + j\omega)(\varepsilon - j\omega)} \right\} \\ &= \lim_{\varepsilon \rightarrow 0} \left\{ \frac{\varepsilon - j\omega}{\varepsilon^2 + \omega^2} \right\} = \lim_{\varepsilon \rightarrow 0} \left\{ \frac{\varepsilon}{\varepsilon^2 + \omega^2} - \frac{j\omega}{\varepsilon^2 + \omega^2} \right\}\end{aligned}$$



First Approach

$\frac{\varepsilon}{\varepsilon^2 + \omega^2}$ is an interesting function, known as the Lorentzian

Its single peak at $\omega = 0$ is $\frac{1}{\varepsilon}$.

It becomes infinitely tall as $\varepsilon \rightarrow 0$.

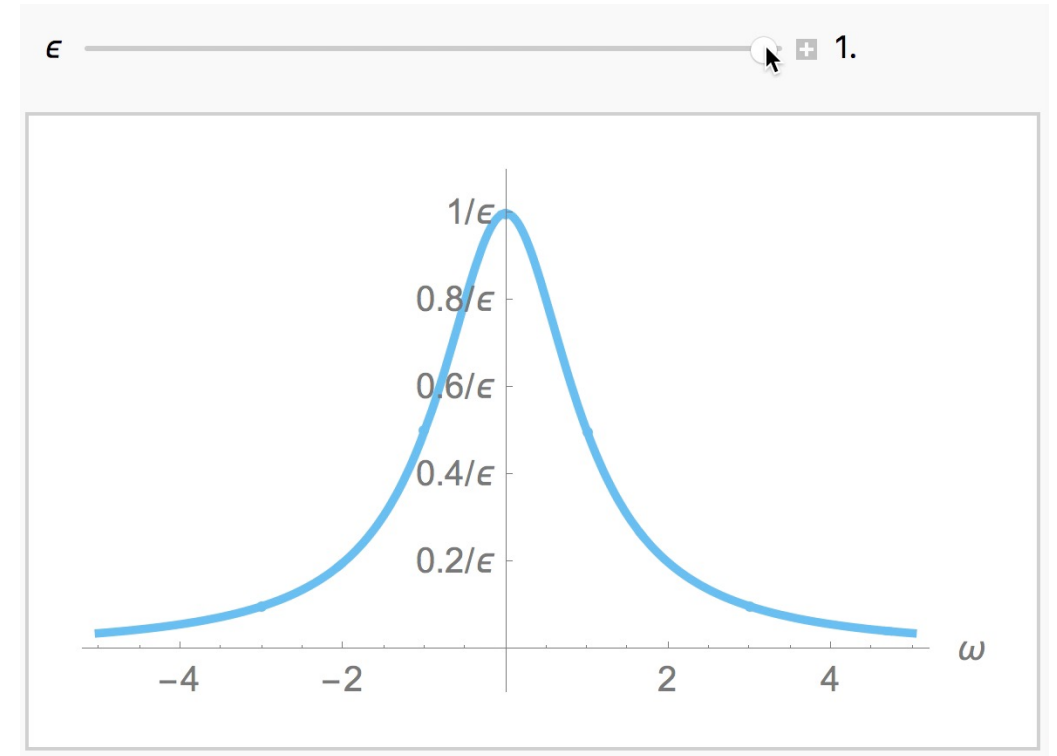
But, the area under the curve is:

$$\int_{-\infty}^{\infty} \frac{\varepsilon}{\varepsilon^2 + \omega^2} d\omega = \tan^{-1} \left(\frac{\omega}{\varepsilon} \right) \Big|_{-\infty}^{\infty} = \pi,$$

regardless of the value of ε

$\Rightarrow$ This function becomes an impulse in the limit as $\varepsilon \rightarrow 0$

$$\lim_{\varepsilon \rightarrow 0} \frac{\varepsilon}{\varepsilon^2 + \omega^2} = \pi \delta(\omega)$$





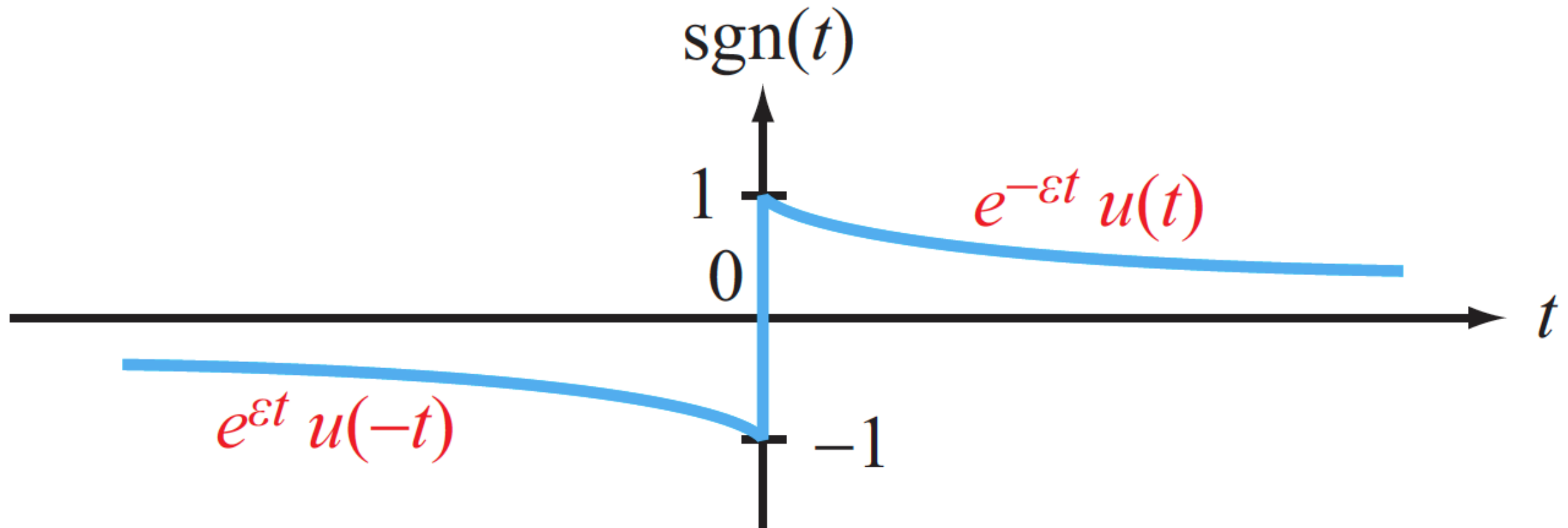
First Approach

$$\begin{aligned}\Rightarrow \mathcal{F}[u(t)] &= \lim_{\varepsilon \rightarrow 0} \left\{ \frac{\varepsilon}{\varepsilon^2 + \omega^2} - \frac{j\omega}{\varepsilon^2 + \omega^2} \right\} \\ &= \lim_{\varepsilon \rightarrow 0} \left\{ \frac{\varepsilon}{\varepsilon^2 + \omega^2} \right\} - \lim_{\varepsilon \rightarrow 0} \left\{ \frac{j\omega}{\varepsilon^2 + \omega^2} \right\} \\ &= \pi\delta(\omega) - \frac{j}{\omega} \\ &= \pi\delta(\omega) + \frac{1}{j\omega}\end{aligned}$$



Second Approach

#2. Write the unit step as the limiting case of two decaying exponentials as $\varepsilon \rightarrow 0$

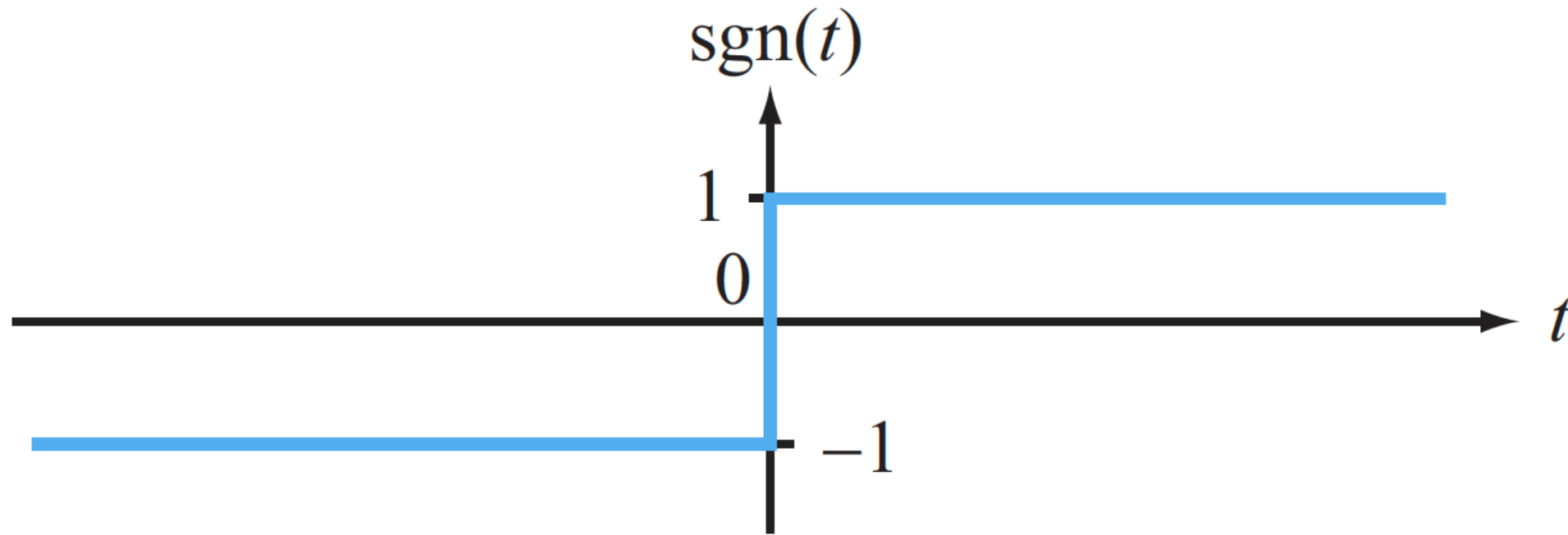


OUR TEXTBOOK PREFERS THIS APPROACH
SINCE THE MATH IS SIMPLER



The Sign Function

The sgn function (formally, the *signum* function), looks like this:



$$\text{sgn}(t) = u(t) - u(-t)$$

$$u(t) = \frac{1}{2}(1 + \text{sgn}(t))$$



Using the Signum function

$$\begin{aligned}\mathcal{F}[\text{sgn}(t)] &= \int_{-\infty}^{\infty} \lim_{\varepsilon \rightarrow 0} \left[e^{-\varepsilon t} u(t) - e^{\varepsilon t} u(-t) \right] e^{-j\omega t} dt \\ &= \lim_{\varepsilon \rightarrow 0} \left[\int_0^{\infty} e^{-(\varepsilon + j\omega)t} dt - \int_{-\infty}^0 e^{(\varepsilon - j\omega)t} dt \right] = \lim_{\varepsilon \rightarrow 0} \left[\frac{e^{-(\varepsilon + j\omega)t}}{-(\varepsilon + j\omega)} \Big|_0^{\infty} - \frac{e^{(\varepsilon - j\omega)t}}{(\varepsilon - j\omega)} \Big|_{-\infty}^0 \right] \\ &= \lim_{\varepsilon \rightarrow 0} \left[\frac{1}{\varepsilon + j\omega} - \frac{1}{\varepsilon - j\omega} \right] = \frac{2}{j\omega}\end{aligned}$$

$$\Rightarrow \mathcal{F}[u(t)] = \mathcal{F}\left[\frac{1}{2}(1 + \text{sgn}(t))\right] = \pi\delta(\omega) + \frac{1}{j\omega}$$

REMEMBER

$$1 \longleftrightarrow 2\pi\delta(\omega).$$



Time Derivative

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{X}(\omega) e^{j\omega t} d\omega$$

Differentiating both sides:

$$\frac{dx(t)}{dt} = \frac{1}{2\pi} \int_{-\infty}^{\infty} j\omega \hat{X}(\omega) e^{j\omega t} d\omega$$

$$\frac{dx(t)}{dt} \leftrightarrow j\omega \hat{X}(\omega)$$

SIMILAR TO LAPLACE TRANSFORM



Time Shift Property

$$\mathcal{F}[x(t - T)] = \int_{-\infty}^{\infty} x(t - T) e^{-j\omega t} dt$$

$$\text{Let } u = t - T \Rightarrow du = dt$$

$$= \int_{-\infty}^{\infty} x(u) e^{-j\omega(u+T)} du = e^{-j\omega T} \int_{-\infty}^{\infty} x(u) e^{-j\omega u} du = e^{-j\omega T} \hat{\mathbf{X}}(\omega)$$

SIMILAR TO LAPLACE TRANSFORM



Frequency Shift Property

$$\begin{aligned}\mathcal{F}\left[e^{j\omega_0 t} x(t)\right] &= \int_{-\infty}^{\infty} x(t) e^{j\omega_0 t} e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} x(t) e^{-j(\omega - \omega_0)t} dt = \hat{\mathbf{X}}(\omega - \omega_0)\end{aligned}$$

$$e^{j\omega_0 t} x(t) \longleftrightarrow \hat{\mathbf{X}}(\omega - \omega_0),$$

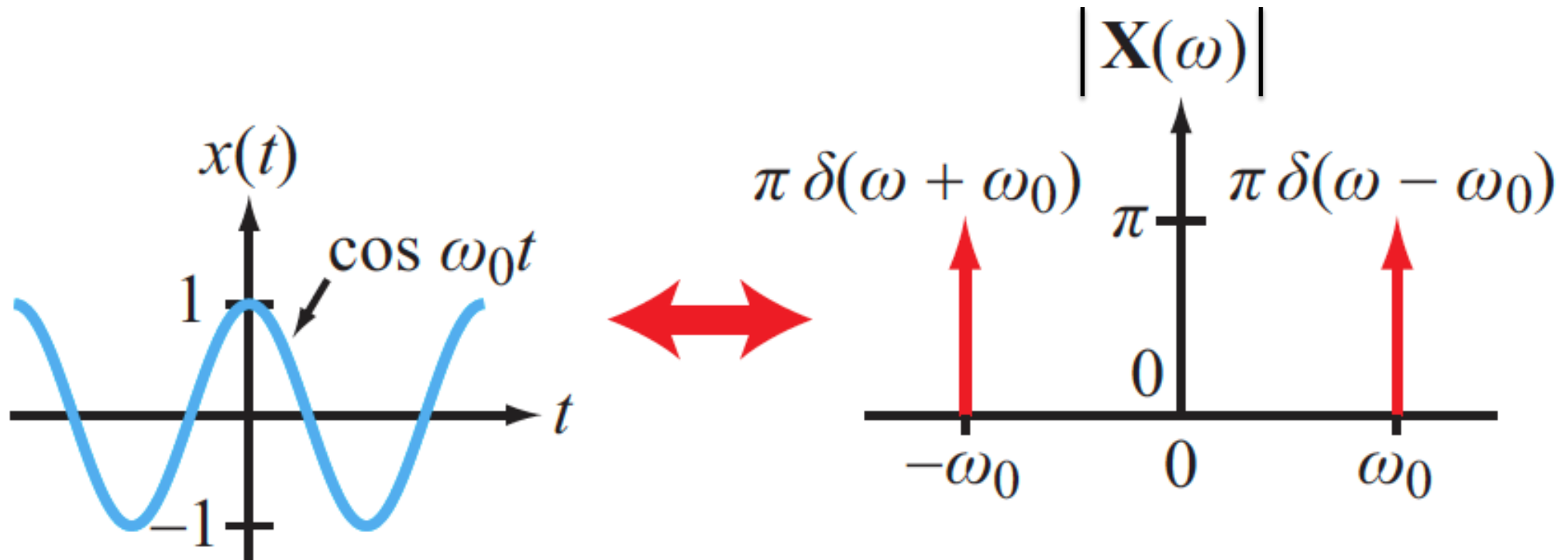
(frequency-shift property)



Fourier Transform of Cosine

$$\cos(\omega_0 t) = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}$$

$$\Rightarrow X(\omega) = \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$

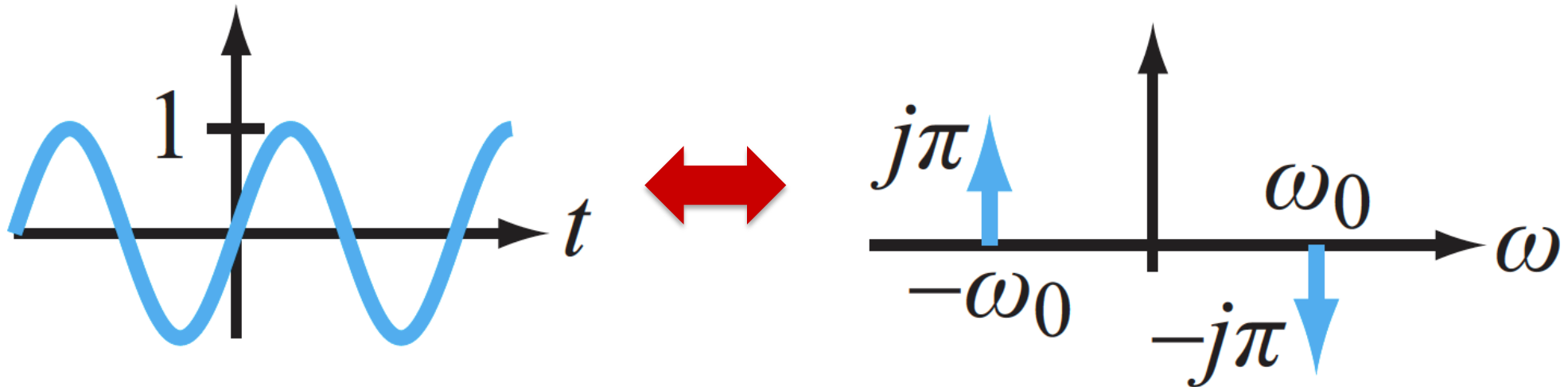




Fourier Transform of Sine

$$\sin(\omega_0 t) = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j} = \frac{-j}{2} (e^{j\omega_0 t} - e^{-j\omega_0 t})$$

$$\Rightarrow X(\omega) = -j\pi\delta(\omega - \omega_0) + j\pi\delta(\omega + \omega_0)$$





Fourier Transform of a Periodic Signal

We know from the exponential form of the Fourier Series that:

$$x(t) = \sum_{n=-\infty}^{\infty} \mathbf{x}_n e^{jn\omega_0 t}$$

Taking the Fourier transform of each term, we get:

$$\mathcal{F}\{x(t)\} = \sum_{n=-\infty}^{\infty} \mathbf{x}_n \mathcal{F}\{e^{jn\omega_0 t}\} = \sum_{n=-\infty}^{\infty} \mathbf{x}_n 2\pi \delta(\omega - n\omega_0).$$

The Fourier transform of a periodic signal is a series of impulses, one impulse for each harmonic (“a pulse train”)



Applications of Frequency Shift

The frequency shift property is the basis for numerous applications in our lives

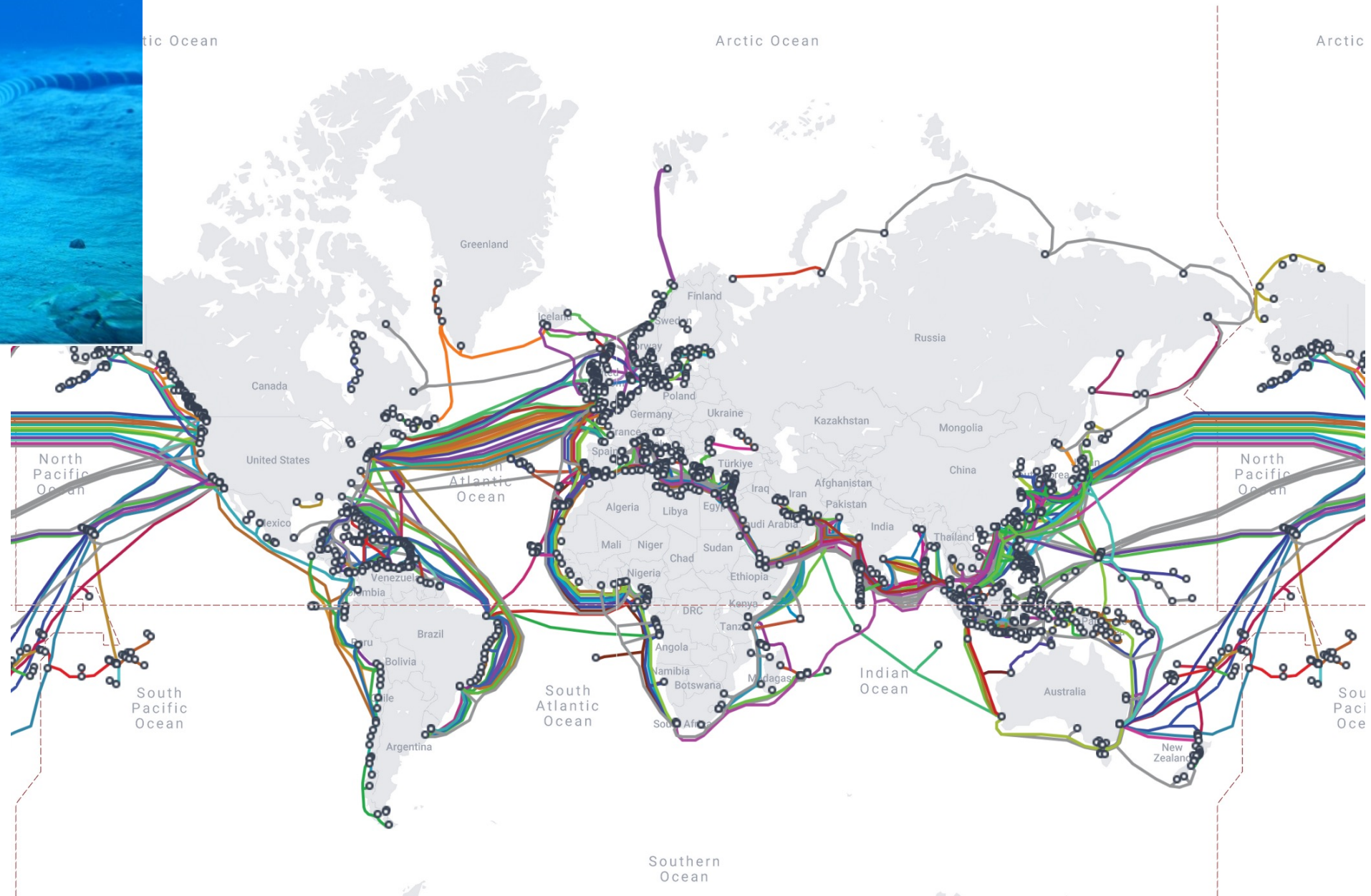
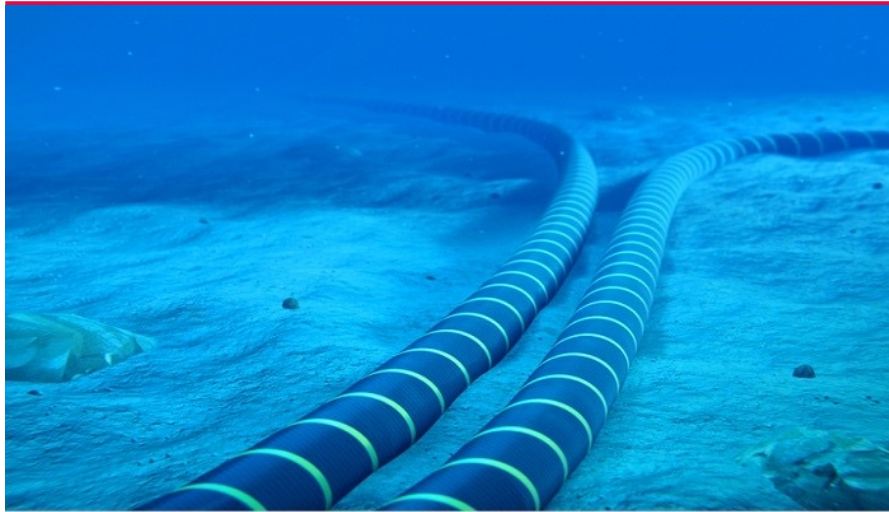
- Signal Modulation
- Radio & Television signal transmission
- Frequency Division Multiplexing (FDM)
 - Many low-bandwidth channels sharing a high bandwidth transmission channel

Transmission channels often operate at a very different band of frequencies compared to the signal that needs to be sent

- Example: Telephone-quality audio frequencies range from 20Hz – 3.5kHz, but an optical fiber line may operate at 1 – 10GHz.

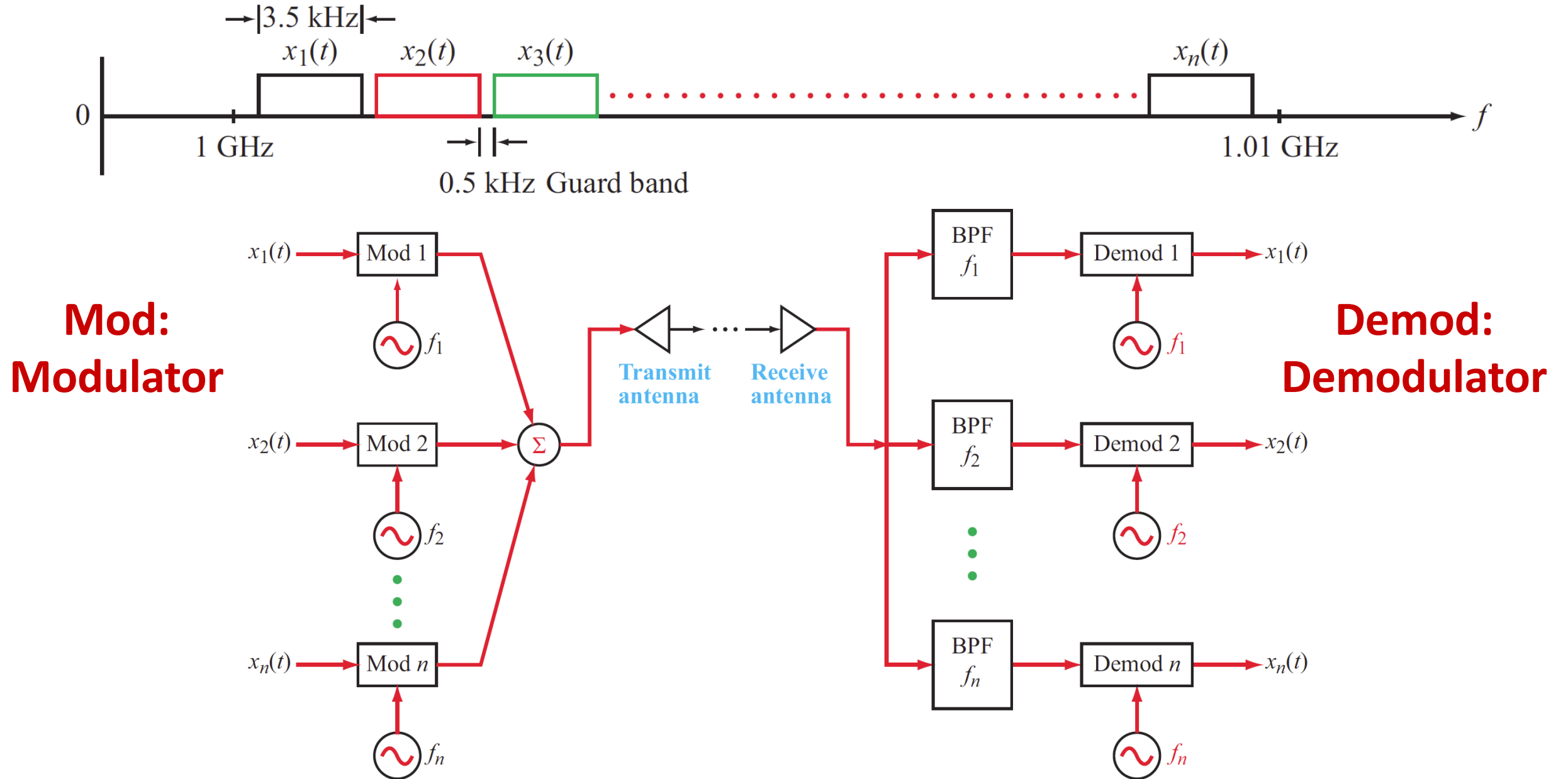


Example: Undersea Data Cables





Frequency Division Multiplexing

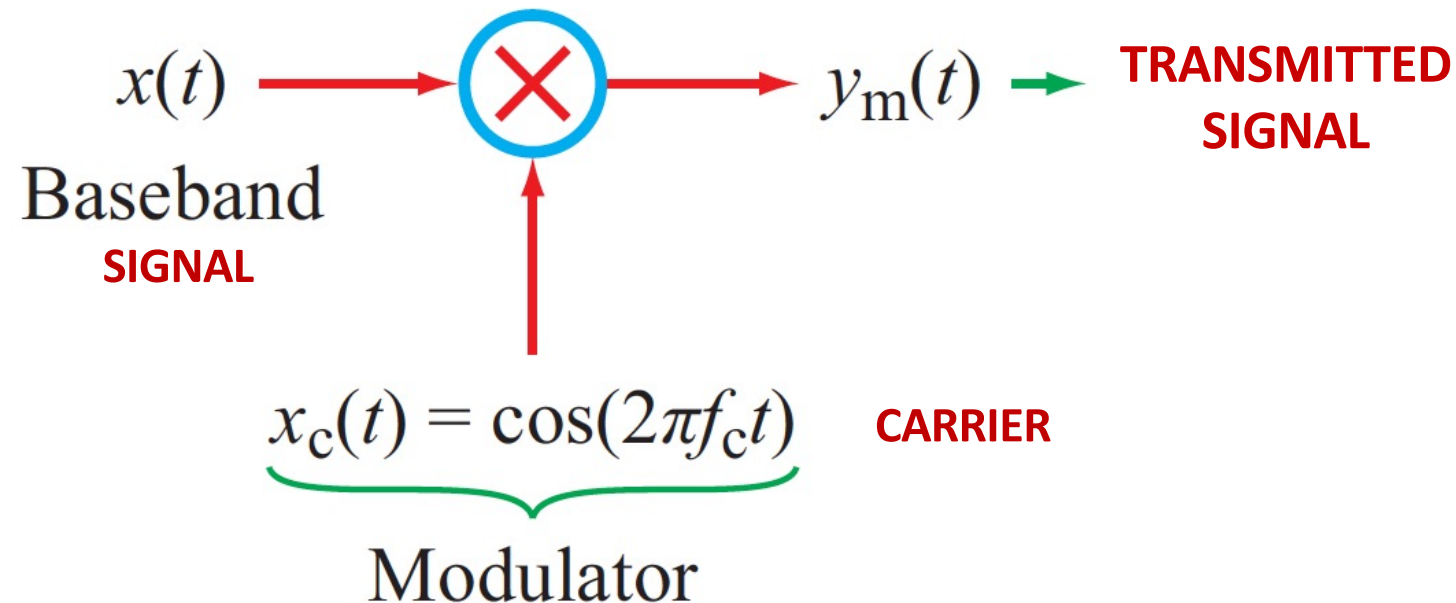




Signal Modulation

Modulation is a powerful concept in signal processing, as we will see below.

- Basically, multiply an information carrying signal $x(t)$ (a.k.a. “baseband”) by a sinusoid (“carrier”)



USUALLY, THE CARRIER'S FREQUENCY IS MUCH HIGHER THAN THE SIGNAL



Fourier Transform of the Modulated Signal

$$\mathcal{F}\{x(t)\cos(\omega_0(t))\} = \mathcal{F}\left\{x(t)\frac{(e^{j\omega_0 t} + e^{-j\omega_0 t})}{2}\right\}$$

$$= \frac{1}{2}\left\{\widehat{\mathbf{X}}(\omega - \omega_0) + \widehat{\mathbf{X}}(\omega + \omega_0)\right\}$$

LOW FREQUENCY TERM

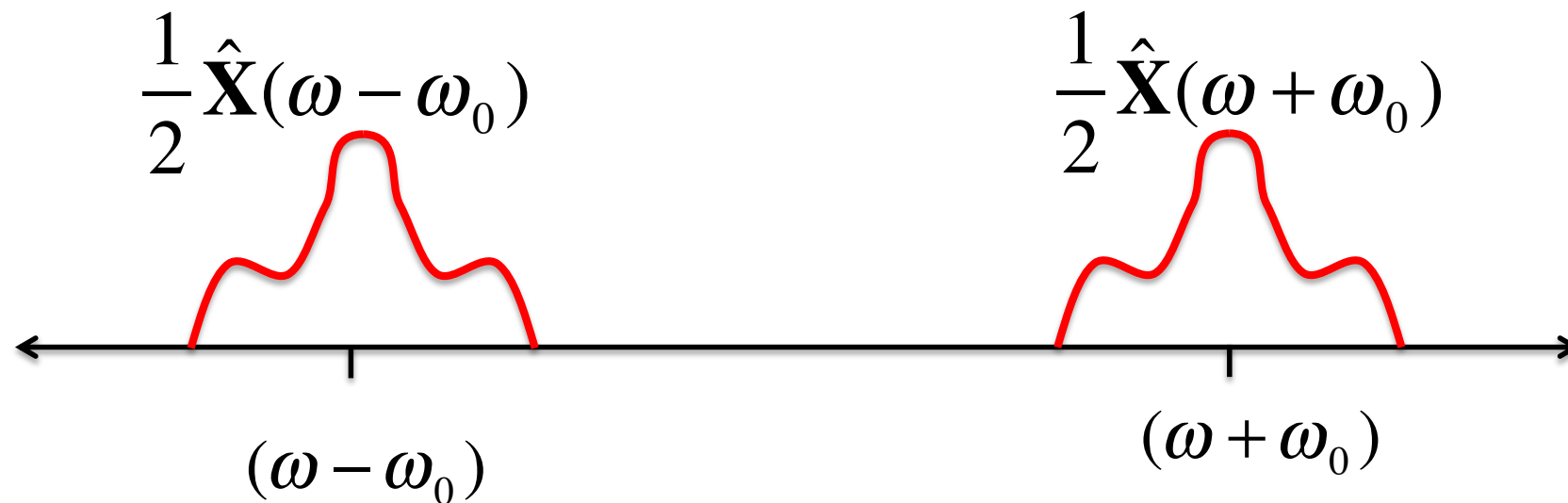
HIGH FREQUENCY TERM



The Spectrum of the Modulated Signal

$$x(t) \cos(\omega_0 t) \longleftrightarrow \frac{1}{2} [\hat{\mathbf{X}}(\omega - \omega_0) + \hat{\mathbf{X}}(\omega + \omega_0)].$$

(modulation property) (5.96)

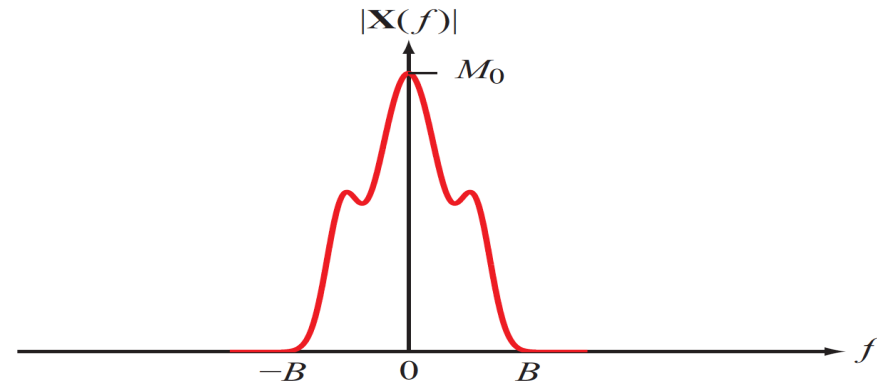
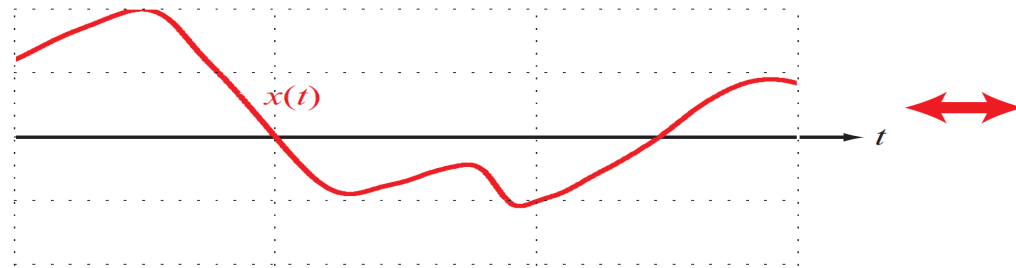


THE CARRIER FREQUENCY IS MUCH HIGHER THAN THE BASEBAND SIGNAL FREQUENCIES

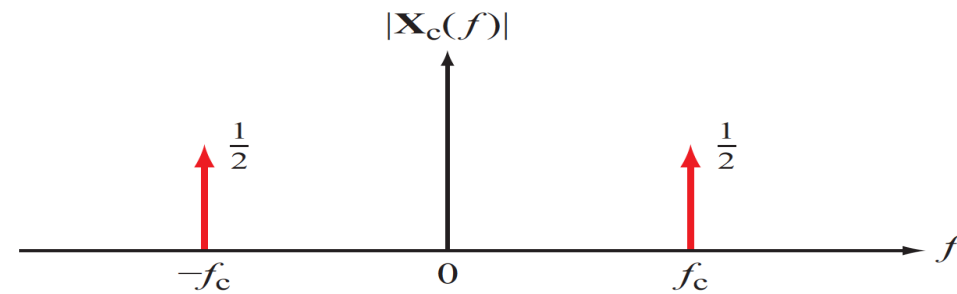
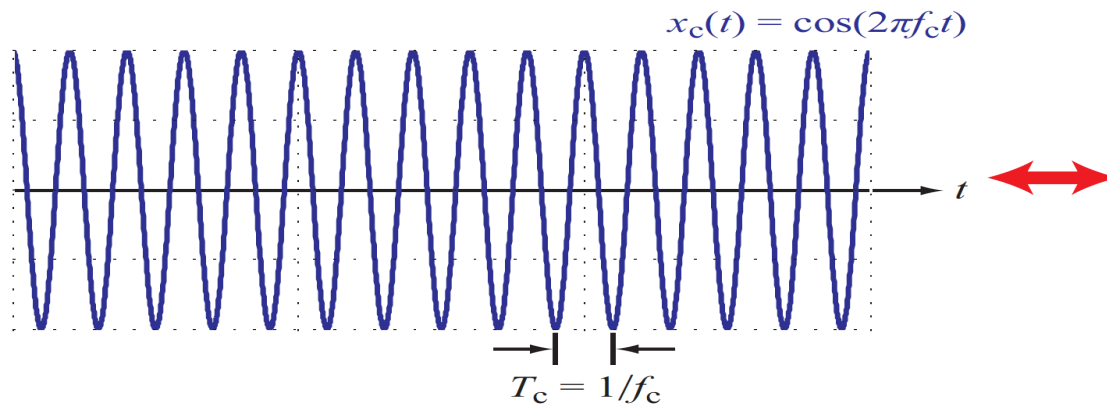
USUALLY, ONE OF THESE BANDS IS KEPT & USED, AND THE OTHER ONE IS FILTERED OUT



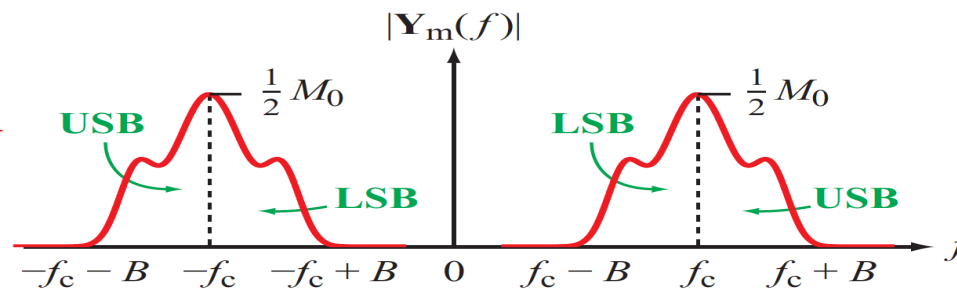
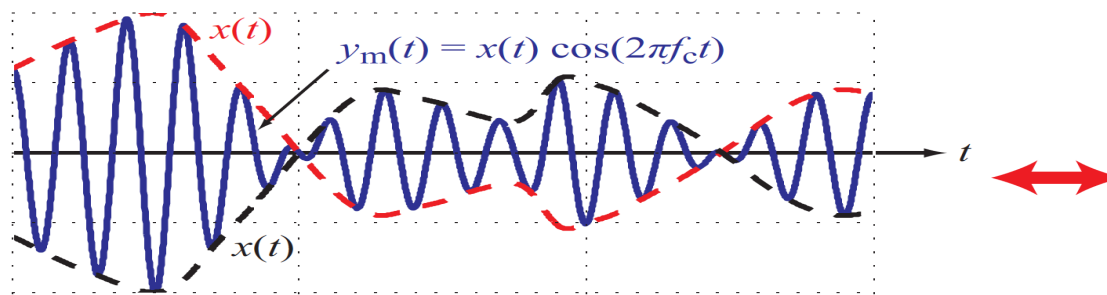
The Spectrum of the Modulated Signal



BASEBAND



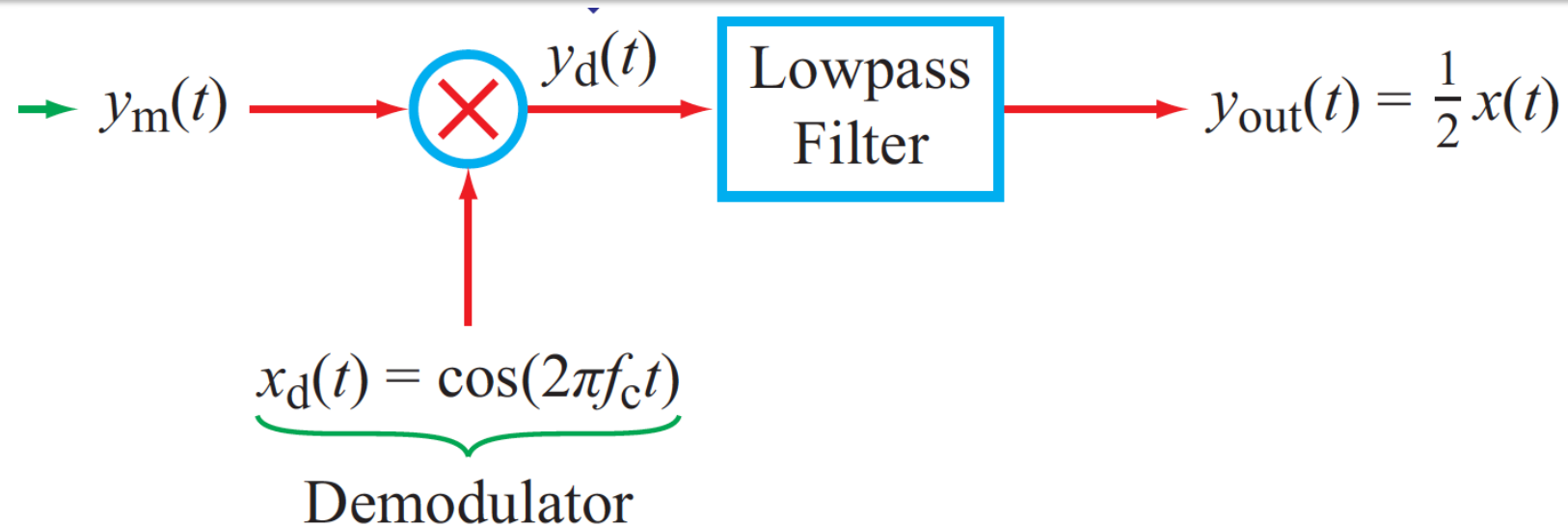
CARRIER



MODULATED CARRIER



Receiving Side



$$y_d(t) = y_m(t) \cos(2\pi f_c t) = x(t) \cos^2(2\pi f_c t)$$

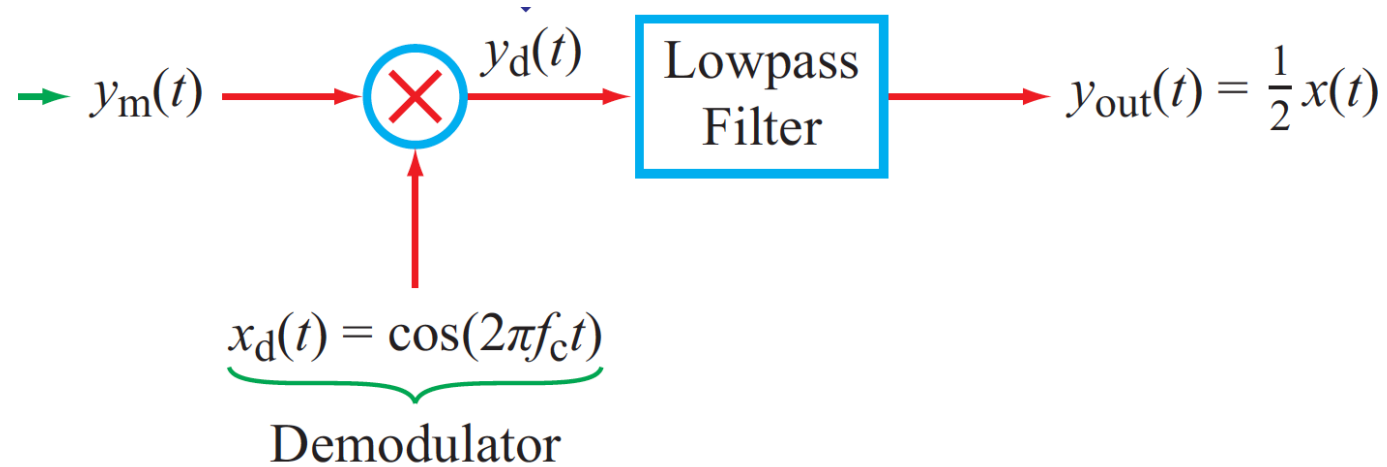
$$= x(t) \frac{[1 + \cos(4\pi f_c t)]}{2}$$

$$= \frac{x(t)}{2} + \cancel{\frac{x(t) \cos(4\pi f_c t)}{2}}$$

USE A LOW-PASS FILTER TO SUPPRESS THIS TERM

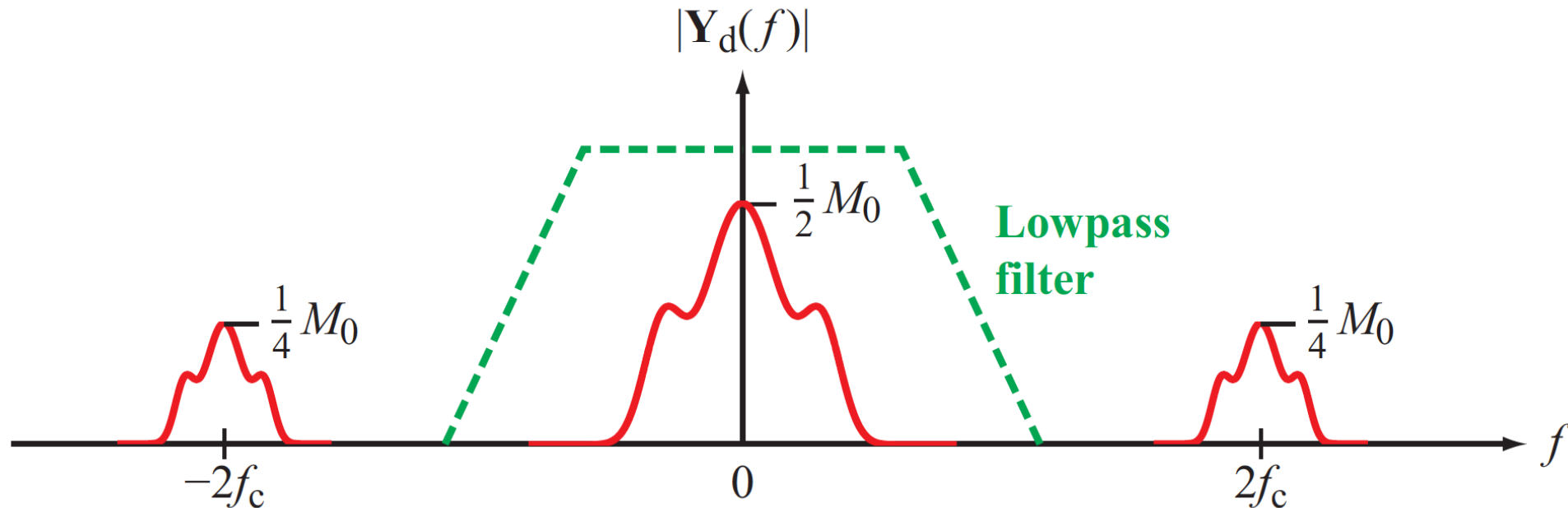


Receiving Side



$$y_d(t) = \frac{x(t)}{2} + \frac{x(t)\cos(4\pi f_c t)}{2}$$

$$Y_d(f) = \frac{X(f)}{2} + \frac{1}{4} [X(f - 2f_c) + X(f + 2f_c)]$$



WE GET BACK THE BASEBAND SIGNAL !



Parseval's Theorem

- The Fourier Transform of a signal captures all of its energy

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\mathbf{X}(\omega)|^2 d\omega$$



Connecting the Dots

Table 5-8: Methods of solution.

Input $x(t)$		Solution Method	Output $y(t)$
Duration	Waveform		
Everlasting	Sinusoid	Phasor Domain	Steady State Component (no transient exists)
Everlasting	Periodic	Phasor Domain and Fourier Series	Steady State Component (no transient exists)
Causal, $x(t) = 0$, for $t < 0$	Any	Laplace Transform (unilateral) (can accommodate non-zero initial conditions)	Complete Solution (transient + steady state)
Everlasting	Any	Bilateral Laplace Transform or Fourier Transform	Complete Solution (transient + steady state)



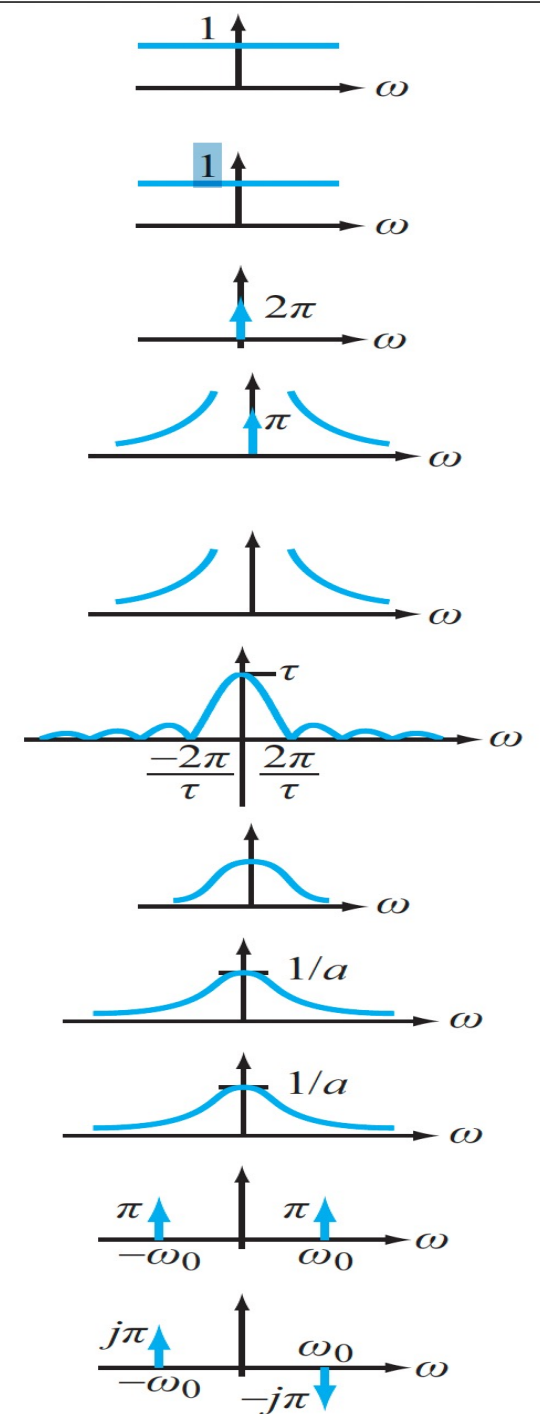
Table 5-7: Major properties of the Fourier transform.

Property	$x(t)$		$\mathbf{X}(\omega) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$
1. Multiplication by a constant	$K x(t)$	$\longleftrightarrow$	$K \mathbf{X}(\omega)$
2. Linearity	$K_1 x_1(t) + K_2 x_2(t)$	$\longleftrightarrow$	$K_1 \mathbf{X}_1(\omega) + K_2 \mathbf{X}_2(\omega)$
3. Time scaling	$x(at)$	$\longleftrightarrow$	$\frac{1}{ a } \mathbf{X}\left(\frac{\omega}{a}\right)$
4. Time shift	$x(t - t_0)$	$\longleftrightarrow$	$e^{-j\omega t_0} \mathbf{X}(\omega)$
5. Frequency shift	$e^{j\omega_0 t} x(t)$	$\longleftrightarrow$	$\mathbf{X}(\omega - \omega_0)$
6. Time 1st derivative	$x' = \frac{dx}{dt}$	$\longleftrightarrow$	$j\omega \mathbf{X}(\omega)$
7. Time n th derivative	$\frac{d^n x}{dt^n}$	$\longleftrightarrow$	$(j\omega)^n \mathbf{X}(\omega)$
8. Time integral	$\int_{-\infty}^t x(\tau) d\tau$	$\longleftrightarrow$	$\frac{\mathbf{X}(\omega)}{j\omega} + \pi \mathbf{X}(0) \delta(\omega)$, where $\mathbf{X}(0) = \int_{-\infty}^{\infty} x(t) dt$
9. Frequency derivative	$t^n x(t)$	$\longleftrightarrow$	$(j)^n \frac{d^n \mathbf{X}(\omega)}{d\omega^n}$
10. Modulation	$x(t) \cos \omega_0 t$	$\longleftrightarrow$	$\frac{1}{2} [\mathbf{X}(\omega - \omega_0) + \mathbf{X}(\omega + \omega_0)]$
11. Convolution in t	$x_1(t) * x_2(t)$	$\longleftrightarrow$	$\mathbf{X}_1(\omega) \mathbf{X}_2(\omega)$
12. Convolution in ω	$x_1(t) x_2(t)$	$\longleftrightarrow$	$\frac{1}{2\pi} \mathbf{X}_1(\omega) * \mathbf{X}_2(\omega)$
13. Conjugate symmetry			$\mathbf{X}(-\omega) = \mathbf{X}^*(\omega)$



BASIC FUNCTIONS

1.		$\delta(t) \longleftrightarrow 1$	
1a.		$\delta(t - t_0) \longleftrightarrow e^{-j\omega t_0}$	
2.		$1 \longleftrightarrow 2\pi \delta(\omega)$	
3.		$u(t) \longleftrightarrow \pi \delta(\omega) + 1/j\omega$	
4.		$\text{sgn}(t) \longleftrightarrow 2/j\omega$	
5.		$\text{rect}(t/\tau) \longleftrightarrow \tau \text{sinc}(\omega\tau/2)$	
6.		$\frac{e^{-t^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}} \longleftrightarrow e^{-\omega^2\sigma^2/2}$	
7a.		$e^{-at} u(t) \longleftrightarrow 1/(a + j\omega)$	
7b.		$e^{at} u(-t) \longleftrightarrow 1/(a - j\omega)$	
8.		$\cos \omega_0 t \longleftrightarrow \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	
9.		$\sin \omega_0 t \longleftrightarrow j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$	

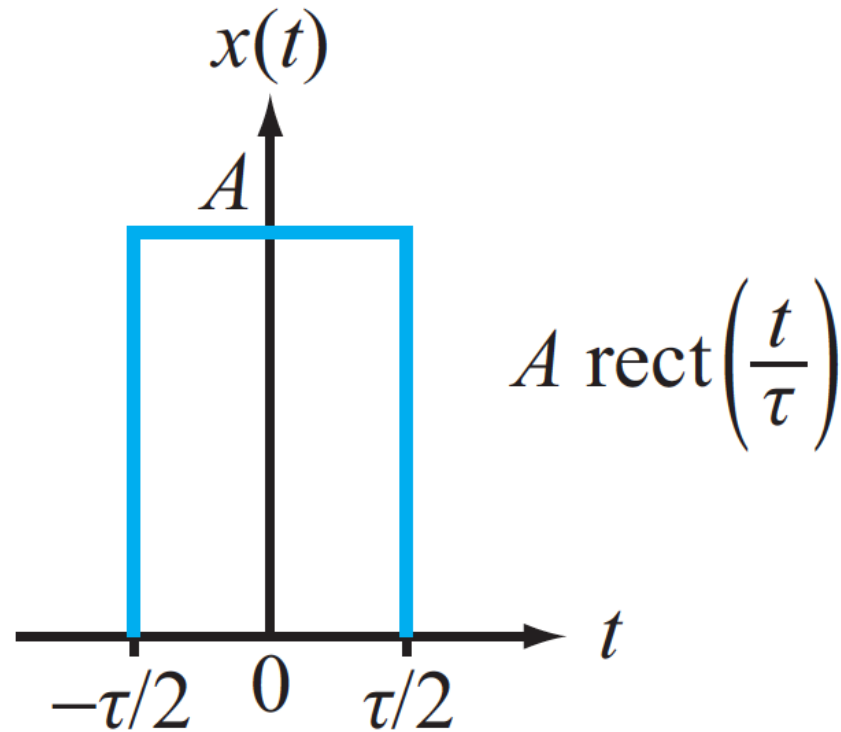




Activity

Re-derive the Fourier transform of a pulse signal by expressing the pulse in terms of the unit step function and using the known Fourier transform for the unit step (from the table)

Signal



CORRECT ANSWER:

$$\mathbf{X}(\omega) = A\tau \operatorname{sinc}\left(\frac{\omega\tau}{2}\right)$$



Summary

- **Today, we looked at the Fourier transform for important signals**
 - The pulse and its relationship with the sinc signal is extremely important
- **We learned how the Fourier transform relates to the Laplace transform**
- **We looked at the key properties of the Fourier transform**
 - Mostly, but not 100% similar to the Laplace
- **We learned a bit about modulation and how an AM radio works**



For the Next Class

- We will cover Sections 5-11 & 5-12 next class
- Work through the textbook examples in Section 5-8
- Study Tables 5.6 and 5.7, and 5.12
- Reflect on Table 5-8 of your Textbook

Table 5-8: Methods of solution.

Input $x(t)$		Solution Method	Output $y(t)$
Duration	Waveform		
Everlasting	Sinusoid	Phasor Domain	Steady State Component (no transient exists)
Everlasting	Periodic	Phasor Domain and Fourier Series	Steady State Component (no transient exists)
Causal, $x(t) = 0$, for $t < 0$	Any	Laplace Transform (unilateral) (can accommodate non-zero initial conditions)	Complete Solution (transient + steady state)
Everlasting	Any	Bilateral Laplace Transform or Fourier Transform	Complete Solution (transient + steady state)